

COMMANDS

```
.control
* === Single sweep: baseline transfer curve ===
alter V_X_P = 1.30
alter V_X_N = 1.20
dc V_W_P 1.15 2.15 0.01
let V_diff_single = v(V_o_p) - v(V_o_n)
plot V_diff_single title "Gilbert Multiplier: Single Weight Sweep"
wrdata /foss/designs/analog-perceptron-gf180mcu/plots/gilbert_single_sweep.txt V_diff_single

setplot const
* === Family of curves, one clean sweep per V_X_P step ===
let n = 0
dowhile n <= 4
    let vxp = 1.10 + n * 0.05
    alter V_X_P = vxp
    dc V_W_P 1.15 2.15 0.01
    let V_diff_fam = v(V_o_p) - v(V_o_n)
    wrdata /foss/designs/analog-perceptron-gf180mcu/plots/gilbert_family_sweep.txt V_diff_fam
    set appendwrite
    let n = n + 1
end
unset appendwrite

alter V_X_P = 1.35
alter V_X_N = 1.2
alter V_W_P = 1.65
.endc
```

MODELS

```
.include $::180MCU_MODELS/design.ngspice
.lib $::180MCU_MODELS/sm141064.ngspice typical
.lib $::180MCU_MODELS/sm141064.ngspice res_typical
* .lib $::180MCU_MODELS/sm141064.ngspice res_statistical
```

